

Experiment 4: Vacuum Cleaner Problem

Aim

To simulate the Vacuum Cleaner problem using Python.

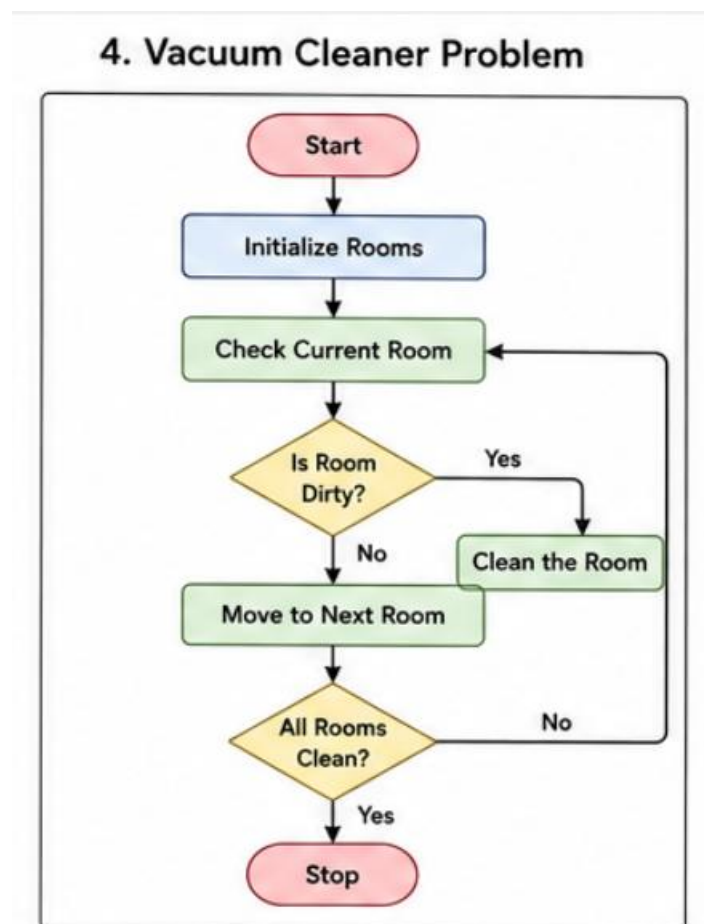
Objective

To clean all dirty rooms automatically.

Algorithm

1. Check each room.
2. If dirty, clean it.
3. Move to the next room.
4. Repeat until all rooms are clean.

Flowchart



Code :

```
rooms = ["Dirty", "Dirty"]
```

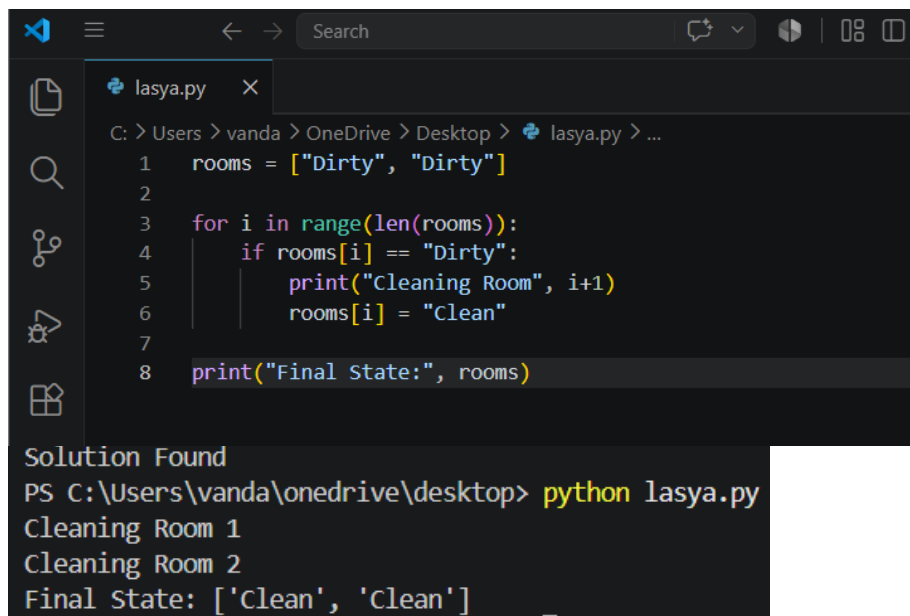
```
for i in range(len(rooms)):
```

```
    if rooms[i] == "Dirty":
```

```
        print("Cleaning Room", i+1)
```

```
        rooms[i] = "Clean"
```

```
print("Final State:", rooms)
```

Output :The image shows a screenshot of a code editor window with a dark theme. The editor has a sidebar on the left with icons for Explorer, Search, Source Control, Run and Debug, and Extensions. The main area shows a file named 'lasya.py' with the following Python code:

```
1 rooms = ["Dirty", "Dirty"]
2
3 for i in range(len(rooms)):
4     if rooms[i] == "Dirty":
5         print("Cleaning Room", i+1)
6         rooms[i] = "Clean"
7
8 print("Final State:", rooms)
```

Below the code editor, there is a terminal window. It shows the command 'python lasya.py' being executed. The output of the script is displayed in the terminal:

```
Solution Found
PS C:\Users\vanda\onedrive\desktop> python lasya.py
Cleaning Room 1
Cleaning Room 2
Final State: ['Clean', 'Clean']
```

Result

All rooms were cleaned successfully.

Conclusion

The Vacuum Cleaner agent cleaned every dirty room automatically.